Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Real-Time Error Prevention System for Industrial Control Operators			
Korea's 12 National Strategic Technology Fields	AI (✓), Advanced Robotics & Manufacturing (✓), Next-gen Communication ( ), Future Mobility ( ), Advanced Bio·Medical (x), Energy·Environment ( ), Quantum ( ), Space·Aviation·Oceanography ( ), Super Convergence ( ), National Defense ( ), Cyber Security ( ), Advanced Materials·Parts ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery & Equipment		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Biomedical Measurement & Monitoring Devices	Industrial Biosignal Monitoring Systems	Software	Embedded Software
Concept of Proposed Technology	Projekt target is an EEG and HRV-based real-time monitoring system designed to enhance industrial safety by detecting operator state changes before errors occur. The system integrates brain activity (EEG) and cardiac rhythm (HRV) measurements to assess operator readiness in real-time. Core Innovation: Rather than detecting fatigue after it occurs (reactive approach), project predicts			

	state degradation 1-5 minutes in advance (predictive approach), allowing time for intervention. Main Functions: • Real-time physiological state monitoring for industrial control room operators • Early warning system for elevated error risk • Integration with industrial control systems (SCADA/PLC) for adaptive automation [10,11] • Operator feedback for self-regulation Potential Applications: • Chemical processing control rooms (prevent hazardous material releases) • Power generation facilities (nuclear, conventional, renewable) • Oil & gas refineries and pipeline operations • Manufacturing process control • Transportation systems (air traffic control, railway operations) Target Market: €12B industrial safety market (2025), growing to €38.5B by 2030 (14.4% CAGR) [5]. Addresses 70-90% of industrial errors attributed to human factors [4].
Rationale for Joint R&D Project	Need for Collaboration: 1. Complementary Expertise: • German Partner: Scientific foundation in neurophysiology, industrial automation protocols, algorithm development • Korean Partner: Hardware miniaturization, wearable device manufacturing, mass production capabilities, Asian market access 2. Hardware Development Requirements: • Integration of medical-grade EEG/HRV sensors into industrial-grade wearable form factor • Korea's strength in consumer electronics and wearable technology manufacturing • Requires expertise in biocompatible materials, low-power electronics, wireless communication 3. Market Access Strategy: • Europe: Strong industrial safety regulations (ATEX, IEC 61508) - German expertise • Asia: Rapidly growing manufacturing sector, large-scale implementation potential - Korean networks • Combined approach enables faster global market penetration

	4. Regulatory Navigation: • Medical device certification (CE, KC) requires local expertise • Industrial automation standards differ by region • Joint approach reduces time-to-market by 12-18 months 5. Technology Transfer: • Scientific algorithms (German partner) + Hardware implementation (Korean partner) • Enables rapid prototyping and field validation in both regions • Shared IP development accelerates commercialization
Key Words	1. EEG-HRV Multi-Modal Integration 2. Real-Time Operator State Monitoring 3. Predictive Error Prevention Systems 4. Edge AI for Industrial Safety 5. SCADA-PLC Integration

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	Primary Objective: Develop and validate a real-time physiological state monitoring system that predicts operator errors in industrial control rooms with 1-5 minute advance warning. Technical Goals: 1. Real-time State Assessment: Develop algorithms to assess operator physiological state with <5 second latency 2. Predictive Analytics: Achieve ≥75% accuracy in detecting elevated error risk 1-5 minutes before incidents [6,7] 3. Industrial Integration: Create communication protocol for SCADA/PLC systems (OPC UA compatible) [10] 4. Wearable Platform: Design industrial-grade form factor for 8-12 hour continuous operation 5. Field Validation: Demonstrate 20-40% error reduction in pilot industrial deployments [4] Scientific Objectives: • Validate correlation between physiological markers and operational errors [1,14] • Establish individualized baseline methodology for inter-operator variability • Develop privacy-preserving data processing architecture (GDPR-compliant) [12]			

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R&D Target Specifications	Hardware Performance: • EEG: Multi-channel dry-electrode system, medical-grade accuracy [8] • HRV: ECG-equivalent precision for cardiac interval measurement [9] • Form Factor: Headband + chest strap, <200g total weight, 8-12hr battery • Environmental: IP54 rating, 0-40°C operation, EMC compliant (EN 61326-1) [13] Software Performance: • State Assessment: Real-time processing with <5 second update interval • Prediction Accuracy: ≥75% sensitivity, ≥85% specificity for error detection [7,15] • Industrial Communication: OPC UA protocol, TLS 1.3 encrypted, <100ms latency [10] • Data Privacy: 100% local processing, no cloud dependency, GDPR-compliant [12] Validation Targets: • Pilot Study: 20-50 operators, 3-6 months monitoring period • Error Reduction: 20-40% decrease in operational errors vs. baseline [4] • False Alarm Rate: <5% (to avoid alarm fatigue) • Operator Acceptance: >70% satisfaction rating for continuous wear comfort
R&D Plan	Phase 1 (Months 1-6): Laboratory Development • Hardware: Select and validate commercial EEG/HRV sensors • Software: Develop baseline data collection system • Integration: Build prototype communication interface for SCADA systems • Milestone: Working laboratory prototype with 2-3 test subjects Phase 2 (Months 7-12): Algorithm Refinement • Data Collection: Gather physiological data from 20-50 operators in controlled settings • Algorithm Development: Optimize state assessment and predictive models • Integration Testing: Validate SCADA/PLC communication in lab environment • Milestone: Validated algorithms with ≥75% prediction accuracy Phase 3 (Months 13-18): Field Validation • Pilot Deployment: Install systems in 1-3 industrial control rooms • Data Collection: 3-6 months continuous monitoring • Safety Analysis: Measure error rate reduction vs. historical baseline • Milestone: Demonstrated 20-40% error reduction in field conditions Phase 4 (Months 19-24): Refinement and Documentation • Hardware Optimization: Finalize industrial-grade form factor with Korean partner

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	• Software Finalization: Implement feedback from pilot users • Documentation: Prepare technical specifications, user manuals, regulatory dossiers • Milestone: Production-ready system, CE/KC certification pathway defined

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1. Scientific Leadership: • Algorithm development for physiological state assessment • Machine learning models for predictive error detection • Statistical validation methodology 2. Industrial Integration Expertise: • Communication protocol specification (SCADA/PLC integration) • Industrial automation standards compliance • Safety system design principles 3. Research Infrastructure: • Laboratory facilities for controlled testing • Access to industrial automation equipment (SCADA simulators, PLC testbeds) • Computational resources for data analysis 4. Intellectual Contributions: • Scientific publication and dissemination • Regulatory compliance strategy (CE marking pathway) • Technology transfer documentation
Expected R&R of Korean SME Partner	1. Hardware Development: • Wearable device design and prototyping • Integration of EEG/HRV sensors into industrial-grade form factor • Manufacturing and quality control for pilot production (100-500 units) 2. Electronics Expertise: • Low-power embedded systems design • Wireless communication implementation (Bluetooth, Wi-Fi) • Battery management and power optimization 3. Field Testing Support:

	• Access to Korean industrial facilities for validation testing • Technical support during pilot deployments • User feedback collection and analysis 4. Commercialization Pathway: • Market analysis for Korean industrial sectors • KC certification and regulatory compliance • Supply chain development and manufacturing scaling 5. Resources: • 2-3 hardware/firmware engineers • Manufacturing facility access • Korean industrial partner network for pilot sites

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Achievements: 1. Novel Monitoring System: First validated implementation of real-time physiological coherence monitoring for industrial error prevention [1,14] 2. Predictive Capability: System achieving ≥75% accuracy in detecting operator state degradation 1-5 minutes before errors [7,15] 3. Industrial Integration: Communication protocol enabling SCADA/PLC systems to receive operator state information in real-time [10,11] 4. Wearable Platform: Industrial-grade device capable of 8-12 hour continuous operation in harsh environments [8,9] Commercialization Potential: Market Size: • Global industrial safety market: €12B (2025) → €38.5B (2030), CAGR 14.4% [5] • Target segments: Chemical (€3.2B), Energy (€2.8B), Oil & Gas (€2.1B), Manufacturing (€1.9B) • Addressable market: ~50,000 control rooms in EU + Korea Revenue Projections: • Years 1-2: €500k-2M (10-50 pilot installations) • Years 3-4: €5-20M (500-2,000 commercial deployments)

	• Years 5+: €50M+ (5,000+ installations, recurring service revenue) Pricing Model: €10,000-50,000 per control room installation (hardware + software + 1-year support) Competitive Advantages: 1. Predictive vs. Reactive: Detects pre-error state (competitors only detect fatigue symptoms after degradation) 2. Scientific Validation: Based on peer-reviewed research with 1.8M+ data points [1] 3. Privacy-Preserving: Local processing, GDPR-compliant (critical for European market) [12] 4. Multi-Modal Integration: Combines cardiac + neural measurements (superior to single-modality systems) [8,15] Broader Impact: Safety Benefits: • 20-40% reduction in human-factor operational errors [4,6] • Estimated €100M-400M annual accident cost savings (Europe alone) [4] • Prevention of catastrophic incidents in chemical, energy, and transportation sectors Regulatory Influence: • Potential adoption as industry standard for safety-critical control room operations [10,11] • Contribution to ISO/IEC standards development for human factors in automation Scientific Contributions: • 3-5 peer-reviewed publications in industrial safety and human factors journals • Open-source dataset of 10,000+ operator-hours physiological data (anonymized) [12] • PhD-level research opportunities at partner institutions Job Creation: • 5-10 direct positions in Korean SME (hardware, software, support) • Technology transfer expertise development (Germany ↔ Korea) • Spin-off opportunities in adjacent markets (aviation, healthcare, defense)

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